IDC101: End-Semester Exam

Name: [redacted]
Roll No: [redacted]

Date : 28 Nov, 2025
Total Marks : 25

Q1 [4 Mark] Write down the output of the following python codes.

(a)
```
def fun(a,b):
    r = 0
    while b > 0:
        r = r + a
        b = b - 1
    return r
print(fun(100,50))
```

(b)
```
l = list(range(15))
print(l[1::3])
print(l[1:-1])
```

(c)
```
import numpy as np
a = np.arange(12).reshape(3,4)
a[:,1] = a[:,1]*2
a[1,:] = a[1,:]*3
print(a)
```

(d)
```
import numpy as np
la = np.arange(0,6,1)
lb = np.vectorize(lambda x: x**2+2)(la)
print(la)
print(lb)
```

Q2 [4 Mark] What is the output from the following program?

```
def fizzbuzz(n):
    for i in range(n):
        s = ""
        if i%3 == 0:
            s = s + "fizz"
        if i%5 == 0:
            s = s + "buzz"
        if s == "":
            s = str(i)
        print(s)
    return
fizzbuzz(16)
```

Q3 **[4 Mark]** Write a python function which accepts a list of items and returns `true` if the list has any duplicate entries and `false` otherwise. Pay attention to properly indenting the code, otherwise no marks will be given.

Q4 **[4 Mark]** What is the output from the following program?

```
def coll(n):
    while n != 1:
        print(n)
        if n%2 == 0:
            n = n // 2
        else:
            n = 3*n + 1
    return n
print(coll(7))
```

Q5 **[4 Mark]** Write a **recursive python function** which prints the same output as what the program in previous question (Q4).

Q6 **[4 Mark]** Write a **recursive python function** which accepts a list of integers, and adds all odd numbers and subtracts all even numbers in the list and returns the final resulting number. For example, applying this function on the list `[1,2,3,4,5,7]` should return `1-2+3-4+5+7 = 10`.

Q7 **[1 Mark]** Write down one advantage and one disadvantage each of bisection and Newton-Raphson methods for finding roots.